

Insect Pest Detection and Identification Method Based on Deep Learning for Realizing a Pest Control System

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Abstract: Detection of insect pests in the agricultural field, which is useful in achieving smart agriculture, has attracted considerable attention. In particular, automatically monitoring the number of crop insect pests has evolved into key means of managing and optimizing agricultural resources. However, despite the fact that conventional convolutional neural network (CNN)-based approaches have yielded sufficient results for general object detection, few methods have been developed to detect and recognize small objects such as insect pests. In addition, no large dataset exists for pest detection even though CNN-based methods require a large dataset to optimize many parameters. In this study, we propose two-stage detection and identification methods for small insect pests based on CNN. We also present a region proposal network for insect pest detection using YOLOv3 and propose a re-identification method using the Xception model. To train these models, we propose a data augmentation method using image processing.

Keywords: image processing, insect pest detection, deep learning

1. INTRODUCTION

Insect pests considerably affect global agricultural economies, as they are known to cause severe damage to crops. Identifying insect pests is critical in agricultural pest forecasting, which is vital for stable agricultural economies and food security. However, identifying insect pests relies heavily on the knowledge of agricultural experts because of the large number of and subtle differences among pest species. Therefore, preventing and controlling insect pests using automatic monitoring systems with camera devices is particularly necessary.

In the field of computer vision, the goal of object detection is not only to determine whether objects from general categories are present in an image, but also to return the spatial location and determine the extent of each object by using a bounding box. Conventionally, traditional methods of detecting insect pests are built on handcrafted features and shallow trainable architectures. However, these methods lack robustness and sensitivity to diversity changes such as object pose, illumination, and part deformation [1–5]. Recently, deep learning techniques [6] have been used as powerful methods to learn feature representations automatically from big data and have considerably improved object detection. Therefore, several deep-learning-based approaches have been proposed [7–12]. In these approaches, convolutional neural networks (CNNs) are used as feature extractors.

Some critical issues must be addressed when implementing insect pest detection using CNN-based detectors. The first is that the sizes of target objects are very small with appearance of intra- and inter-class objects being highly similar. Because a CNN is

designed to learn automatically and adaptively the spatial-feature hierarchies layer by layer, sub-sampling layers in a feature hierarchy lead to an inherent multiscale pyramid. As a result, producing feature maps at different spatial resolutions is required. Therefore, the higher layers are the most robust to variations such as object pose, illumination, and part deformation because they have large receptive fields and strong semantics. One critical disadvantage is that geometric details are lost and resolution is low. If target objects are small (e.g., insect pests), critical features for detecting the object may not appear in subsequent layers. In other words, small object detection is quite challenging for existing CNN-based object detection methods. The second critical issue is the dataset for training a CNN model, where a large-scale dataset is required. Few practical and large datasets currently exist for insect pest detection, as using cameras to capture numerous insects in real field environments is costly. In addition, most insect pest images currently in public datasets are collected in controlled lab environments, which do not effectively satisfy real-world requirements of insect pest recognition in the actual field environments.

In this study, we propose a two-stage detection and identification methods for small insect pests based on CNN. For insect pest detection, we focus on the You Only Look Once (YOLO) v3, which is a state-of-the-art and real-time object detection method [13]. YOLOv3 is a single-shot object detector and is trained end to end from raw pixels to ultimate categories. In general, when the size of a target object is relatively large, YOLOv3 can detect the object with high accuracy. By contrast, when the size of the target object is small, the object often goes undetected or over-detected by YOLOv3. To address this issue, during the first stage of our method, it

detects some regions where insect pests may be located by using YOLOv3 with a low threshold of detection. In other words, YOLOv3 is used as a region proposal network. During the second stage of our method, all detected region proposals are re-identified by a CNN-based classifier. We used the Xception model [14] as a CNN-based classifier because it is a state-of-the-art CNN that yields high computational efficiency and outperforms other CNN models. As a final contribution, we propose a novel data augmentation method using image processing to train these models.

2. RELATED WORK

In early stages of development for insect pest recognition, handcrafted-feature-based methods were the primary solutions. Mayo et al. proposed an automatic identification method using support vector machines (SVMs) without manually specifying the region of interest [1]. Samanta et al. proposed a diagnosis system of tea insect pests by correlation-based feature selection and artificial neural networks [2]. Yalcin proposed an insect classification method by employing four feature extraction methods such as Hu moments and local binary patterns [3]. Xie et al. proposed an insect recognition system by combining a sparse-coding technique for encoding insect images with multiple-kernel learning [4]. Rani et al. proposed a SVM classifier to identify whiteflies, aphids, and thrips in leaf images [5]. However, considerable time and technical knowledge are required to perform manual and complicated image processing as part of preprocessing for these methods.

These handcrafted-feature-based methods tend to perform evaluations based on extremely small datasets with few categories, as designing feature extractors for recognizing diverse insect pests is both inefficient and time-consuming. Most importantly, using a handcrafted-feature-based approach to represent high-level semantic information is difficult.

However, the visual recognition paradigm changed rapidly after the emergence of the ImageNet dataset, which demonstrated the power of data-driven feature learning. During the past years, CNN has become the most effective architecture for visual recognition. Hence, many studies on deep-learning-based approaches for insect pest detection and identification have been conducted. Lim et al. adopted Alexnet and Softmax to build an insect classification system [7]. In their study, the effects of classification performance on the change in the number of kernels in the convolution layer and on different image datasets were examined. Rustia et al. proposed an identification and detection method for multi-class insects using yellow sticky paper by employing cascaded CNNs [8]. Shen et al. proposed a detection and identification method using a faster Regions with CNN (R-CNN) [15] with an improved Inception-V3 [16] for stored-grain insects under field conditions with impurities [9]. Xia et al. proposed a detection and identification method based on an improved VGG19 [17] to detect insect pests quickly and

accurately [10]. Bhatt et al. focused on accuracy performance of existing typical meta-architectures such as the faster R-CNN [18] and Single Shot multibox Detector (SSD) [19] for insect pest detection [11]. He et al. proposed an approach to detect oilseed rape pests based on deep learning [12]. In their study, the effectiveness of typical object detection algorithms in oilseed rape pest detection was evaluated.

These conventional methods have some disadvantages in terms of practicality, where insects are detected only when a very simple background such as a supplementary sheet is used. However, in real field environments, the size of an insect pests in an image can be relatively small. In addition, some regions similar in appearance to that of the insect pests may appear in the background. Moreover, deep-feature-based studies were forced to employ an insufficient number of samples for optimizing the numerous hyper-parameters of CNNs due to the fact that a large-scale dataset was lacking.

3. MATERIALS AND METHODS

3.1 Overview

The objective of this study is to detect precisely small insect pests from an image captured in a real field environment. We focused on *Phyllotreta striolata* and *Phyllotreta atra* as insect pest species for detection. Fig. 1 shows a sample image with both species on a cruciferae plant. Both species were approximately 2 mm long, and their appearances were very similar. The only difference in appearance was the yellow stripes on the back of one (and thus absent on the other). In addition, the backgrounds in the images captured for both species were similar, including the appearance of gaps between leaves or leaf damage due to the consumption behavior of the insect pests.

The flow of the proposed method is shown in Fig. 2. During the first stage, the proposed method detects some region proposals from an input image by YOLOv3 [13] with a low threshold of detection. Thus, YOLOv3 is used as a region proposal network. In the second stage, all region proposals are re-identified using the Xception model. For insect pests that are extremely difficult to detect and identify, the second stage of our approach suppresses candidates that are intentionally overdetected during the first stage.

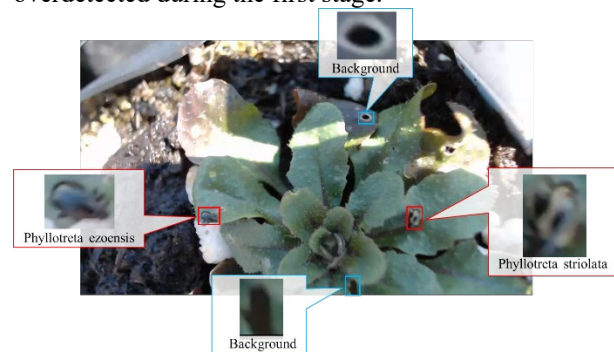


Fig. 1 Example of a captured image in a real field environment.

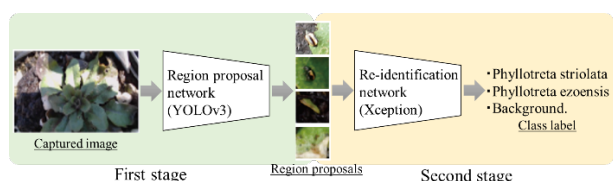


Fig. 2 Flow of the proposed two-stage detection method.

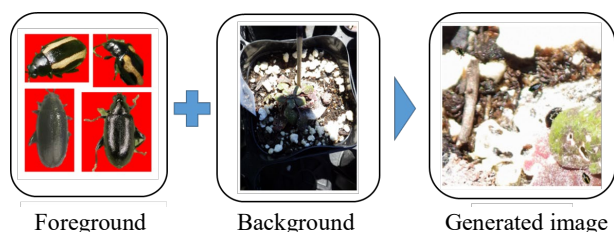


Fig.3 Example of data augmentation for the first stage of the proposed method (using YOLOv3).

3.2 Dataset and data augmentation

Although optimizing many parameters of YOLOv3 based on deep learning requires a large dataset, only small datasets currently exist for pest detection and identification. Existing datasets have only approximately 100 images for each type [2, 8, 10–12, 20–23]. As a result of our collection *P. striolata* and *P. atra* of images from the Internet, the number of high-resolution images for use as learning data was only approximately 30 for each species.

Data augmentation for training a CNN-based model, including cropping, flipping, scaling, rotating, adding noise, and color transformation, is typically performed as part of preprocessing a dataset. Increasing the number of training samples is useful for reducing overfitting and improving generalization.

In this study, we propose augmented datasets for two-stage detection. First, we propose data augmentation to train effectively YOLO v3 for use as a region proposal network by pasting real segmented objects into natural images. The flow of our data augmentation is shown in Fig. 3. A mask image as a foreground image is created by extracting only insect pest regions from collected pest images. A single image is then randomly selected from a large natural image dataset that does not contain insect pests. A background image is cropped at 512×512 pixels from the selected image. Here, image processing operations such as cropping, flipping, and scaling are conducted to the fore- and background images. Then, by combining a foreground image with a background image, we construct a large training image dataset. Note that overlapping foreground images are allowed when combining multiple foreground images into a single background image. When the images are combined, additional image processing operations such as adding Gaussian noise, gamma transform, and Gaussian blur are conducted to diversify the training data.



Fig. 4 Example of data augmentation for the second stage (using Xception) (left: *P. striolata*, center: *P. atra*, right: background).

Second, we propose data augmentation to train the Xception model used as a classifier. The basic concept here is the same as the data extension for YOLO, which combines a masked foreground image with a background image that does not contain insect pests. Here, some image processing operations such as cropping, flipping, and scaling are conducted to the fore- and background images. For the combined image, additional image processing operations such as adding Gaussian noise, gamma transform, and Gaussian blur are conducted. Note that at most one foreground image is composited. Fig. 4 shows an example of data augmentation for the second stage of the proposed augmentation method.

3.3 Pest candidate detection using a CNN-based object detector

With the rapid development of deep learning, effective and efficient CNN-based object detection models have been proposed, including faster R-CNN, Region-based Fully Convolutional Networks (R-FCN) [24], You Only Look Once (YOLO) [25], and SSD. Two-stage detectors such as faster R-CNN and R-FCN perform detection with relatively high accuracy but have unrealistic computational costs. By contrast, one-stage detectors such as YOLO and SSD have lower accuracy but faster speed.

Redmon et al. proposed YOLO for unified detector casting object detection as a regression problem from image pixels to spatially separated bounding boxes and associated class probabilities [25]. Because the region proposal generation stage is completely dropped, unlike in region-based approaches, YOLO directly detects objects using a small set of candidate regions. In particular, YOLO divides an image into an $S \times S$ grid, from which C class probabilities, B bounding box locations, and confidence scores are predicted.

Redmon et al. proposed an improved version of YOLO as YOLOv3 [13]. In YOLOv3, the CNN-based feature extractor is replaced by the Darknet-53 feature extractor. YOLOv3 has achieved state-of-the-art results on standard detection tasks. However, YOLOv3 may fail to detect small objects, possibly because of the coarse grid division and because each grid cell can only contain a few objects.

To address this issue, during the first stage of the proposed method, YOLOv3 is used as a region proposal network. A threshold for judging insect pest candidate regions, which is one of the parameters of YOLOv3, is set to relatively low. The aim of our approach is to detect many small insect pest areas, with some

overdetection.

3.4 Pest identification using CNN-based classifier

In the second stage of the proposed method, as shown in Fig. 2, region proposals detected by YOLOv3 are re-identified as insect pests or not. We employ the Xception model as a classifier during this stage.

Xception (meaning “extreme inception”) is an extension of the inception architecture that replaces the standard inception modules with depth-wise separable convolutions [14]. Depth-wise separable convolutions consist of a depth-wise convolution (a spatial convolution performed independently for each channel), followed by a pointwise convolution (1×1 convolution across channels). This architecture idea is based on the inception module hypothesis, which separates cross-channel and spatial correlations to simplify and make the convolution process more efficient.

Because the size of the region proposal depends on the output of YOLOv3, an input image for Xception is resized to the required size of Xception while ignoring the aspect ratio. To suppress the overdetection background region as well as re-identify the insect pests, the proposed Xception has three corresponding output classes: *P. striolata*, *P. atra*, and background. A Softmax function is used for the output layer of the re-identification task, as this task is a multi-class classification problem. As a loss function, we used standard Softmax cross-entropy.

4. EXPERIMENTS AND RESULTS

4.1 Experimental setups

To demonstrate the effectiveness of the proposed two-stage insect pest detection method, we compared our method to the conventional technique. For the conventional technique, we focused on a case in which only YOLOv3 was used for detection.

We employed a Logicoool C920r camera, where the camera was positioned sufficiently close without escaping insect pests from plants. Insect pests located on cruciferous plants grown outdoors were photographed in video mode. The number of video frames used in the experiment was 474, and the total number of insect pests contained in them was 646 (*P. striolata*: 427, *P. atra*: 219).

We next describe inputting the captured frames to YOLOv3. We chose 512×512 pixels as an input image size for YOLO. When the image was resized to 512×512 pixels (where the frame size of the original captured image was $1,920 \times 1,080$ pixels), an insect pest with a body length of only approximately 2 mm appeared considerably compressed such that not only YOLOv3 but also human experts were unable to identify it. To address this issue, when inputting captured frames into YOLOv3, we adopted a sliding window approach. As shown in Fig. 5, each frame was cut into eight separate images, which were 800×800 pixels with overlap. Each cropped image was then resized to 512×512 pixels, which was the required input size for YOLOv3.

We prepared 12,000 images using the proposed data

augmentation method for training YOLOv3, where 80% of the images were used for training and the remaining 20% for validation. YOLOv3 was trained in 100 epochs. The batch size was set to 16.

We next describe inputting Xception during the second stage of detection. Xception is used for re-identifying the region proposal outputted from YOLOv3. The region proposal was resized to 229×229 pixels while ignoring the aspect ratio.

We prepared 4,000 images for each category using the proposed data augmentation method for training Xception, where 80% of the images were used for training and the remaining 20% for validation. Xception was trained by Adagrad in 100 epochs. The dropout rate for each task was set to 0.5. The batch size was set to 16.

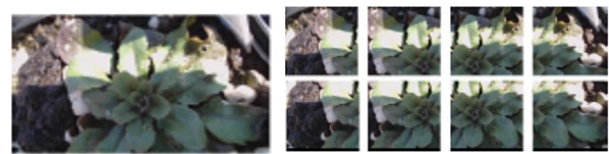
4.2 Results and discussion

Confusion matrices by the two methods are shown in Tables 1 and 2. In our experiment, an IoU value of 0.5 was used to accept ground truth objects as predicted in detection tasks. The confusion matrices of the results in the first stage using only YOLOv3 and in the second stage using the two-stage process are shown in Tables 1 and 2. In these tables, the rows and columns of the confusion matrix correspond to the true classes and predicted classes, respectively.

In the case of using only YOLOv3 for detection, although YOLOv3 outputted 1,316 as insect pest regions, 78% of the output represented overdetection. This result was reasonable because a threshold for judging insect pest candidate regions is set to relatively low for detecting as many small insect pests as possible. If suppressing overdetection is desired, it can be achieved by setting a relatively high threshold. However, the undetected rate will be even worse than 65%.

However, after re-identification using Xception, many erroneously detected background areas were then correctly identified as background classes. We confirmed that considerable overdetection in the first stage was suppressed in the second stage. In addition, as one merit of the proposed method, most of the misclassification of *P. atra* as *P. striolata* in the first stage were correctly re-identified as *P. atra* in the second stage.

Results of precision, recall, and f-measure for the two methods are shown in Table 3. The proposed two-stage detection was compared with one-stage detection using only YOLOv3. Recall was nearly the same, but precision was greatly improved. We thus confirmed that the f value was significantly improved.



(a) Captured frame (b) Eight cropped images
Fig. 5 Sliding window approach for input to YOLOv3.

Table 1 Confusion matrix of one-stage detection using only YOLOv3

	<i>P. striolata</i>	<i>P. atra</i>	Background
<i>P. striolata</i>	124	7	0
<i>P. atra</i>	62	97	0
Background	576	450	0

Table 2 Confusion matrix of two-stage detection using YOLOv3 and Xception

	<i>P. striolata</i>	<i>P. atra</i>	Background
<i>P. striolata</i>	130	1	0
<i>P. atra</i>	13	108	38
Background	115	132	779

Table 3 Comparison of one- and two-stage detection

	One-stage	Two-stage
Precision	0.16	0.77
Recall	0.34	0.36
F-measure	0.21	0.50

4. CONCLUSION

In this study, we proposed a method to detect and identify insect pests from images using deep learning to support pest control systems in agriculture. For detecting insect pests that are small and similar in appearance, YOLOv3, which is a state-of-the-art CNN-based object detection method, was used as a region proposal network. Xception, which is a state-of-the-art CNN-based object classification model, was then used for re-identification of several region proposals outputted from YOLOv3. By using the proposed two-stage detection method, we confirmed that undetected insect pests and background over-detection were suppressed.

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